HALDEN

Pumpen GmbH - Aalen, Germany

Kreiselpumpe / Centrifugal Pump

HCP-350

Installation, Operation and Maintenance Manual

Translation of the original instructions (German)

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Controller firmware	3.x

Read this manual before installation and commissioning. Keep it at the place of installation for the full service life of the pump.

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1 General information

1.1 About this manual

This manual applies to Halden HCP-350 end-suction centrifugal pumps with serial numbers from HCP350-210001, supplied as a bare-shaft pump or as a pump unit on a common baseplate with motor and Halden PC-3 pump controller. It is part of the product. Personnel responsible for installation, operation and maintenance must have read and understood it.

Information on accessories not supplied by Halden (motors, variable frequency drives, couplings of other makes) is given in the respective manufacturer's documentation. Where that documentation contradicts this manual for the pump itself, this manual applies.

Halden publishes technical changes that affect safety or reliability as Service Bulletins between manual revisions. Service Bulletins take precedence over this manual for the units and serial ranges they name. Current bulletins are available from your Halden service partner.

1.2 Symbols and signal words

DANGER	Immediate hazard. Will result in death or serious injury if not avoided.
WARNING	Possible hazard. Could result in death or serious injury if not avoided.
CAUTION	Possible hazard. Could result in minor injury or damage to the pump or the installation.
NOTE	Information for efficient and trouble-free operation.

1.3 Manufacturer

Halden Pumpen GmbH, Industriestrasse 14, 73430 Aalen, Germany. Service hotline +49 7361 555 0140 (Monday to Friday, 07:00 - 17:00 CET). For North American installations, contact your authorised Halden service partner.

1.4 Warranty

Halden warrants the pump against defects in material and workmanship for 24 months from delivery or 18 months from commissioning, whichever expires first. The warranty does not cover wear parts (mechanical seals, wear rings, bearings, gaskets), damage caused by dry running, cavitation, operation outside the permitted operating range, use of non-original spare parts, or failure to carry out the maintenance described in chapter 8. Claims must be submitted through the authorised service partner with the serial number and the controller event log.

2 Safety

2.1 Intended use

The HCP-350 is intended for pumping clean or slightly contaminated water and water-glycol mixtures up to 90 C and a maximum working pressure of 16 bar. Pumping of flammable, toxic or abrasive liquids requires written approval from Halden. The pump is not designed for use in potentially explosive atmospheres and carries no ATEX marking.

2.2 Personnel

Installation, commissioning and maintenance may only be carried out by trained specialists who are familiar with the relevant regulations and this manual. Electrical work on the motor and controller must be carried out by a qualified electrician.

2.3 Residual risks

WARNING	Rotating shaft and coupling. Never operate the pump with the coupling guard removed.
WARNING	Hot surfaces. Pump casing and bearing housings can exceed 70 C during operation. Allow to cool before maintenance.
CAUTION	Dry running destroys the mechanical seal within seconds. Never start the pump unless the casing is completely filled.

2.4 Before working on the pump

1. Switch off the motor and secure it against restart (lockout/tagout).
2. Close the suction and discharge isolation valves.
3. Relieve pressure from the casing through the drain plug and wait until the pump has cooled.
4. Check that no pressure remains on the casing gauge connection before opening any joint.

3 Transport and storage

Lift the pump unit only at the lifting lugs on the baseplate. Do not lift by the motor eyebolt, which is rated for the motor alone. Weight of the complete unit: approximately 640 kg.

If the pump is stored for more than three months before commissioning, store it indoors in a dry place, keep flange covers fitted, and turn the shaft by hand by a few revolutions once a month to prevent brinelling of the bearings. After storage longer than twelve months, replace the bearing grease before commissioning.

4 Product description

4.1 Design

The HCP-350 is an end-suction, single-stage pump of back pull-out design. The motor, coupling spacer and bearing frame can be withdrawn with the impeller as one assembly without disturbing the piping. The shaft is sealed with a single cartridge-type mechanical seal.

Each pump is supplied with a Halden PC-3 controller monitoring suction and discharge pressure (transmitters PT-1 and PT-2), bearing temperature, seal leak detection, motor current, and the differential pressure across the suction strainer (gauges PI-01 upstream and PI-02 downstream of the strainer).

4.2 Type key

Example: HCP-350-310-55-4 = pump size / impeller diameter in mm / motor power in kW / number of motor poles.

4.3 Technical data

Parameter	Value
Rated flow / head	350 m3/h at 60 m
NPSH required at rated point	4.1 m
Motor	55 kW, 4-pole, 1780 rpm at 60 Hz (1480 rpm at 50 Hz)
Max. working pressure / temperature	16 bar / 90 C
Flanges suction / discharge	DN 150 / DN 125
Impeller diameter (standard)	310 mm
Impeller-to-casing (wear ring) clearance	0.40 - 0.55 mm, diametral
Casing bolt tightening torque	45 N-m, cross pattern, two passes (lubricated threads)
Bearings	6311 C3 (drive end) / 3311 (non-drive end)
Grease quantity per bearing	18 g, lithium complex NLGI 2

Parameter	Value
Mechanical seal kit	18-0355
Weight, complete unit	approx. 640 kg
Sound pressure level at 1 m	79 dB(A)

5 Installation

5.1 Foundation and baseplate

The foundation must be rigid enough to absorb vibration and have a mass of at least three times the mass of the pump unit. Level the baseplate to within 0.2 mm/m using shims next to the foundation bolts, then grout it with non-shrink grout. Do not tighten the foundation bolts fully until the grout has cured (typically 48 hours).

5.2 Piping

Support the piping independently so that no forces or moments are transmitted to the pump flanges. Connect the flanges only after the grout has cured. The suction line should be as short and straight as possible, with a straight length of at least five pipe diameters before the pump inlet, and must rise continuously toward the pump to prevent air pockets. Use an eccentric reducer with the flat side on top.

CAUTION Do not install a throttling valve in the suction line. The suction isolation valve must be fully open during operation (see 7.2).

A suction strainer (basket strainer, mesh 3 mm, Halden part 18-1120) must be installed between the suction isolation valve and the pump inlet, with pressure gauges PI-01 and PI-02 on either side so that the strainer differential pressure can be read.

5.3 Coupling alignment

Align the coupling after the piping has been connected. Permitted misalignment: radial offset 0.05 mm, angular 0.05 mm per 100 mm. Use a laser alignment tool where available. Re-check alignment after 24 hours of operation at operating temperature.

5.4 Electrical connection

Connect the motor according to the motor nameplate and the wiring diagram in the controller door. Connect transmitters PT-1 and PT-2 to terminals X3.1 to X3.4 with shielded cable, shield earthed at the controller end only. The PC-3 provides Modbus TCP on port 502 for connection to the plant control system (see chapter 11).

5.5 Permitted nozzle loads

Forces and moments from the piping must not exceed the values below (EN ISO 5199, cast iron casing). Higher loads distort the casing, change the wear ring clearance and shorten bearing and seal life.

Flange	Fx (N)	Fy (N)	Fz (N)	Mx (N-m)	My (N-m)	Mz (N-m)
Suction	2,350	2,100	2,650	1,050	750	860
Discharge	1,600	1,450	1,800	700	500	580

6 Commissioning

1. Check that the suction strainer basket is installed and clean.
2. Open the suction isolation valve fully. Leave the discharge valve closed.
3. Fill the pump and the suction line with liquid. Vent the casing through the vent plug until liquid without air bubbles emerges.
4. Check the direction of rotation by switching the motor on briefly. The direction must match the arrow on the bearing housing.
5. Start the pump. Open the discharge valve slowly within 30 seconds of start-up. Never run against a closed discharge valve for more than 30 seconds.
6. Set the controller limits for suction pressure and strainer differential pressure (factory default: strainer dP alarm at 0.25 bar).
7. Record suction and discharge pressure, motor current, bearing temperature and vibration at both bearings on the commissioning record. Keep the record with this manual.

7 Operation

7.1 Operating range

Operate the pump between 40% and 110% of the flow at best efficiency point (BEP). Continuous operation below 25% of BEP flow causes recirculation, temperature rise and vibration and is not permitted.

7.2 Suction conditions

The NPSH available at the pump inlet must exceed the NPSH required (4.1 m at the rated point) by a margin of at least 1.0 m. NPSH available falls when the suction strainer clogs, when the suction isolation valve is partly closed, when the liquid level in the supply tank drops, or when the liquid temperature rises. Insufficient NPSH causes cavitation: a crackling noise like gravel in the pump, fluctuating discharge pressure and, over time, pitting damage to the impeller inlet.

7.3 Controller display

The PC-3 shows the operating values on the main screen and active faults as a P-code with a short text. Faults are latched until acknowledged. Pressing RESET acknowledges the fault once its cause has cleared. The last 200 events are stored in the event log (Menu > Service > Event log) and can be exported to USB.

8 Maintenance

8.1 Maintenance schedule

ID	Task	Interval	Reference
M-01	Grease pump and motor bearings	2,000 operating hours	8.2
M-02	Check coupling alignment	4,000 h or 12 months	5.3
M-03	Inspect mechanical seal(s) for leakage	4,000 h	8.4
M-04	Clean suction strainer	750 h, or when the strainer dP alarm is active	8.3
M-05	Measure vibration at both bearings	1,000 h	9.2
M-06	Measure impeller-to-casing clearance	8,000 h	8.5
M-07	Check foundation bolt torque	8,000 h	5.1
M-08	Calibrate PT-1 / PT-2 transmitters	24 months	4.1

NOTE Intervals apply to normal duty (clean water, up to 6,000 operating hours per year). Shorten them for contaminated liquids or continuous duty.

8.2 Lubrication

Grease each bearing with 18 g of lithium complex grease NLGI 2 through the grease nipple while the pump is running. Open the grease relief plug during greasing and close it after five minutes. Over-greasing raises bearing temperature and is the most frequent cause of fault P-02 after maintenance.

8.3 Cleaning the suction strainer (task M-04)

1. Stop the pump and secure it against restart.
2. Close the suction isolation valve and the valve between strainer and pump inlet.
3. Relieve pressure through the strainer drain and remove the strainer cover.
4. Lift out the basket, clean it with water and a soft brush, and check the mesh for tears. Replace damaged baskets (part 18-1120).
5. Refit the basket and cover with a new cover gasket. Open the valves, vent, and restart.
6. Confirm that the strainer differential pressure (PI-01 minus PI-02) has returned to below 0.05 bar.

8.4 Replacing the mechanical seal

1. Isolate, drain and cool the pump (see 2.4). Remove the coupling guard and the coupling spacer.
2. Remove the casing bolts and withdraw the back pull-out assembly using the jacking screws.
3. Remove the impeller nut (left-hand thread) and pull off the impeller. Withdraw the seal cartridge.
4. Fit a new seal kit (18-0355). Lubricate O-rings with water or glycerine only, never with oil or grease.
5. Refit the impeller and the back pull-out assembly with a new casing gasket. Tighten the casing bolts to 45 N-m in a cross pattern, in two passes.
6. Refit the coupling spacer, re-check alignment, and refit the guard.

CAUTION Do not exceed the specified casing bolt torque. Over-tightening can crack the gasket seat in the casing.

8.5 Measuring the impeller-to-casing clearance (task M-06)

Measure the diametral clearance between impeller wear ring and casing wear ring with a feeler gauge at four points, 90 degrees apart. Permitted clearance: 0.40 - 0.55 mm. If the clearance exceeds the upper limit, replace the casing wear rings. Pump efficiency falls by roughly 1 to 2% for every 0.1 mm above the upper limit.

8.6 Replacing the bearings

1. Remove the rotating assembly as described in 8.4, steps 1 to 2.
2. Remove the bearing covers and pull the bearings off the shaft with a puller acting on the inner ring only.
3. Heat the new bearings to 90 - 110 C with an induction heater (never with an open flame) and slide them onto the shaft against the shoulder.
4. Pack each bearing with 18 g of grease and refit the covers with new lip seals.
5. Reassemble and re-check alignment (5.3). Measure bearing temperature and vibration after the first 4 hours of operation.

8.7 Tightening torques

Unless stated otherwise, torques apply to lightly oiled threads, property class 8.8. Tighten in a cross pattern in at least two passes.

Joint	Thread	Torque (N-m)
Casing bolts (casing to cover)	M16	45

Joint	Thread	Torque (N-m)
Bearing housing to casing	M12	60
Bearing covers	M10	25
Strainer cover	M12	45
Pump feet to baseplate	M20	180
Foundation bolts	M24	310
Impeller nut	M30 LH	120

9 Faults and troubleshooting

9.1 Controller fault codes

The following faults are displayed by the PC-3 controller. Check the first measure listed before replacing any component.

Code	Text	Trigger	Possible causes	First measure
P-01	Suction pressure low	PT-1 below setpoint for 30 s	Supply tank level low; suction valve closed; strainer blocked	Check tank level and suction valve position
P-02	Bearing temp. high	Bearing RTD above 85 C	Over-greasing after maintenance; lack of grease; misalignment	Check grease record. Do not add grease if M-01 was done in the last 500 h
P-03	Vibration high	Overall velocity above 7.1 mm/s	Misalignment; loose foundation bolts; impeller imbalance	Check foundation bolt torque, then coupling alignment
P-04	Seal leakage	Leak sensor wet for 60 s	Worn seal faces; seal damage after dry running	Inspect seal flush line; replace seal if leakage exceeds 10 drops/min
P-05	Motor overload	Current above 110% FLA for 20 s	Liquid viscosity/density change; impeller rubbing; supply voltage imbalance	Compare the three phase currents; check impeller clearance (8.5)
P-06	Dry running	No flow with motor running for 15 s	Empty supply tank; air lock; pump not primed	Stop pump. Vent casing and refill before restart
P-07	Cavitation	Suction pressure fluctuation above 15% for 10 s	NPSH available too low: strainer clogged, suction valve throttled, liquid temperature high	Read the strainer dP at PI-01/PI-02. If above 0.25 bar, clean the strainer (8.3)
P-08	Sensor fault	PT-1 or PT-2 signal outside 4-20 mA	Wire break; moisture in connector; transmitter failure	Check wiring at X3.1 - X3.4 and connector seals

9.2 Troubleshooting by symptom

Symptom	Possible cause	Remedy
Pump delivers too little	Wrong direction of rotation; worn wear rings; air in suction line	Check rotation; measure clearance (8.5); check suction line for leaks
Pump delivers nothing	Not primed; suction valve closed; strainer completely blocked	Prime and vent; open valve; clean strainer
Noise like gravel in the pump	Cavitation	See P-07
Vibration rising over weeks	Bearing wear; impeller fouling or damage	Trend vibration; inspect bearings and impeller
Motor overheats	Operation far right of BEP; voltage too low	Throttle discharge to the rated point; check supply
Seal leaks soon after replacement	O-rings damaged during fitting; dry running at start-up	Replace seal; always vent before start

Vibration evaluation: overall velocity below 2.8 mm/s is good, 2.8 to 7.1 mm/s is acceptable for continuous operation but should be trended, above 7.1 mm/s requires action (fault P-03). A peak at running speed (1x) usually indicates imbalance; at twice running speed (2x), misalignment; at blade-pass frequency, hydraulic causes such as operation far from BEP.

10 Spare parts

Order spare parts through your authorised Halden service partner, stating the pump serial number. Recommended spares for two years of operation are marked with *.

Item	Part no.	Description	Qty
101	10-3501	Casing	1
102	10-3502	Casing cover	1
230	12-3500	Impeller, 310 mm	1
502*	14-3505	Casing wear ring	1
433*	18-0355	Mechanical seal kit (seal, O-rings, gaskets)	1
400*	16-3500	Casing gasket	1
321*	17-6311	Bearing, drive end	1
322*	17-3311	Bearing, non-drive end	1
160	18-1120	Suction strainer basket, 3 mm mesh	1
161*	18-1121	Strainer cover gasket	2
710	19-0301	Pressure transmitter PT-1 / PT-2, 0-16 bar, 4-20 mA	1

11 Modbus register map

PC-3 controller, Modbus TCP, port 502, unit ID 1. Holding registers, 16-bit, big-endian. Read-only unless marked R/W.

Register	Name	Unit / scaling	Access
40001	Suction pressure PT-1	bar x 100	R
40002	Discharge pressure PT-2	bar x 100	R
40003	Strainer differential pressure	bar x 100	R
40004	Bearing temperature DE	C x 10	R
40005	Bearing temperature NDE	C x 10	R
40006	Motor current	A x 10	R
40007	Vibration overall DE	mm/s x 100	R
40010	Active fault code	1-8 = P-01 to P-08, 0 = none	R
40011	Operating hours	h	R
40020	Strainer dP alarm setpoint	bar x 100	R/W
40021	Suction pressure alarm setpoint	bar x 100	R/W
40030	Remote start/stop	0 = stop, 1 = start	R/W

12 Revision history

Rev.	Date	Change
A	2021-03-01	First edition
B	2021-11-15	Section 11: Modbus register map added
C	2022-06-09	Section 10: seal kit 18-0355 replaces 18-0350
D	2023-05-22	Section 9: faults P-07 (cavitation) and P-08 (sensor fault) added; controller firmware 3.x

13 Decommissioning and disposal

For shutdowns longer than four weeks, drain the pump completely, dry it with compressed air, and turn the shaft by hand once a month. Where frost is possible, the pump must be drained even for short shutdowns. Before restarting after a long shutdown, carry out the commissioning steps in chapter 6.

At the end of its service life, drain and clean the pump. Separate the cast iron, steel, copper (motor) and electronic (controller) parts and dispose of them through authorised recycling companies. Bearing grease and contaminated cleaning materials are hazardous waste in most jurisdictions.

Annex A Commissioning record

Item	Value	Item	Value
Pump serial no.		Date	
Site / tag		Commissioned by	
Suction pressure (bar)		Discharge pressure (bar)	
Flow (m3/h)		Motor current L1/L2/L3 (A)	
Bearing temp. DE / NDE (C)		Vibration DE / NDE (mm/s)	
Strainer dP (bar)		Direction of rotation checked	
Alignment radial / angular (mm)		Signature	

Annex B Maintenance record

Date	Op. hours	Task ID	Findings / parts used	Name

Annex C EC Declaration of Conformity

Translation of the original declaration.

The manufacturer, Halden Pumpen GmbH, Industriestrasse 14, 73430 Aalen, Germany, declares under its sole responsibility that the product: centrifugal pump HCP-350, serial numbers from HCP350-210001, complies with the relevant provisions of: Machinery Directive 2006/42/EC; Low Voltage Directive 2014/35/EU (pump units with controller); EMC Directive 2014/30/EU (pump units with controller).

Harmonised standards applied: EN 809:1998+A1:2009, EN ISO 12100:2010, EN 60204-1:2018.

Person authorised to compile the technical file: Dipl.-Ing. M. Kessler, Head of Engineering, at the address above.

Aalen, 2023-05-22

Dr. A. Brandt, Managing Director